Electronic Devices & Circuits Laboratory (EE2301)

Experiment 8: NMOS Linear Voltage Regulator using a 741 Op-Amp

1.2 V Regulated Output from a 3 to 5 V Supply

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1 Objective

The aim of this experiment is to:

1. Build a linear voltage regulator using an LM741 op-amp driving the gate of an N-channel MOSFET connected as a source follower.
2. Keep the output voltage steady at $V_{OUT} = 1.200\,\text{V}$ while the input supply varies from $3.0\,\text{V}$ to $5.0\,\text{V}$.
3. Measure **line regulation** (mV/V) and see how the feedback adjusts the gate voltage to compensate for channel-length modulation in the MOSFET.
4. Measure **load regulation** across load currents up to about $118\,\text{mA}$ and extract the closed-loop output resistance R_{OUT}.
5. Find the **dropout voltage** at full load ($R_L \approx 10.2\,\Omega$) and relate it to the MOSFET's on-resistance $R_{DS(\text{on})}$.
6. Measure power dissipation and efficiency, and understand why a linear regulator's efficiency is limited to V_{OUT}/V_{IN}.
7. Check feedback stability on an oscilloscope with three different output capacitors ($10\,\mu\text{F}, 47\,\mu\text{F}, 94\,\mu\text{F}$) and check for oscillation.
8. Answer all eleven analysis questions from Section 9 of the lab manual.

2 Theory

2.1 How the Feedback Loop Works

The regulator uses negative feedback to keep V_{OUT} equal to a fixed reference $V_{REF} = 1.200\,\text{V}$. The LM741 op-amp acts as the error amplifier:

- Non-inverting input (Pin 3) is connected to $V_{REF} = 1.200\,\text{V}$.
- Inverting input (Pin 2) is directly connected to V_{OUT} (source of M_1).
- Op-amp output (Pin 6) drives the gate of the N-channel MOSFET.

The MOSFET is in source-follower (common-drain) configuration, so:

$$V_{OUT} = V_G - V_{GS} \tag{1}$$

If V_{OUT} drops (e.g. load increases or V_{IN} decreases), the op-amp sees a positive error ($V_{REF} - V_{OUT}$) and drives the gate higher. This increases V_{GS}, so the MOSFET passes more current and V_{OUT} rises back to V_{REF}. If V_{OUT} rises, the op-amp lowers V_G and reduces current.

With unity feedback ($\beta = 1$) and open-loop gain $A \approx 2 \times 10^5$:

$$V_{OUT} = V_{REF}\left(\frac{T}{1+T}\right), \quad \text{Steady-state error} = \frac{V_{REF}}{1+T} \approx \frac{1.200\,\text{V}}{2 \times 10^5} \approx 6\,\mu\text{V} \tag{2}$$

So ideally, the output stays within $6\,\mu$V of the reference.

2.2 Line Regulation and Output Resistance (Small-Signal)

Using small-signal analysis with the MOSFET in the feedback loop:

Line Sensitivity (dV_{OUT}/dV_{IN}): Writing KCL at the output node with $v_g = -Av_{out}$:

$$\frac{dV_{OUT}}{dV_{IN}} = \frac{1}{r_o\left[g_m(1+A) + \frac{1}{r_o} + \frac{1}{R_L}\right]} \approx \frac{1}{r_o g_m (1+A)} \tag{3}$$

With $A \approx 2 \times 10^5$, $g_m \approx 0.85\,\text{A/V}$, $r_o \approx 50\,\Omega$:

$$\frac{dV_{OUT}}{dV_{IN}} \approx \frac{1}{50 \times 0.85 \times 2 \times 10^5} \approx 1.18 \times 10^{-7}\,\text{V/V} = 0.00012\,\text{mV/V} \tag{4}$$

This is too small to measure directly with a bench multimeter (0.1 mV resolution).

Closed-Loop Output Resistance (R_{OUT}):

$$R_{OUT} = \frac{1}{g_m(1+A) + \frac{1}{r_o}} \approx \frac{1}{g_m A} \approx \frac{1}{0.85 \times 2 \times 10^5} \approx 5.88\,\mu\Omega \tag{5}$$

In practice, breadboard contact and wire resistance (a few mΩ) dominates, so measured R_{OUT} will be much larger.

2.3 Why the LM741 Needs Separate ± 12 V Supplies

The 741 cannot run from the same 3–5 V input rail for two reasons:

1. **Output swing:** The 741's output stays about 2 V below its positive rail. At $V^+ = 3.0$ V, the gate voltage would be limited to ≈ 1.0 V, but we need $V_G \approx 2.1$–2.2 V to keep $V_{OUT} = 1.2$ V.
2. **Input common-mode range:** The 741 also cannot sense voltages within 2 V of its negative rail (ground), so it cannot sense the 1.2 V output.

So we power the op-amp from ± 12 V (or ± 10 V) auxiliary supplies.

2.4 Why a Bleeder Resistor is Needed

An NMOS source follower can only *source* current — it cannot pull the output down. Without a bleeder resistor:

- When the load is disconnected, there is no path for the output to discharge.
- Leakage or gate capacitance can charge C_{OUT} and pull V_{OUT} above V_{REF}.

A 1.2 kΩ bleeder draws a steady 1 mA and provides a discharge path:

$$I_{bleeder} = \frac{V_{OUT}}{R_{bleeder}} = \frac{1.200\,\text{V}}{1.2\,\text{k}\Omega} = 1.00\,\text{mA} \tag{6}$$

2.5 Dropout Voltage

When V_{IN} drops close to V_{OUT}, the drain-to-source voltage V_{DS} becomes very small. The op-amp saturates high to drive the gate as high as possible, and the MOSFET enters the ohmic (triode) region, behaving like a small resistor $R_{DS(\text{on})}$. The output then follows:

$$V_{OUT} = V_{IN}\left(\frac{R_L}{R_L + R_{DS(\text{on})}}\right) \tag{7}$$

Dropout is defined (per the manual) as the V_{IN} at which V_{OUT} falls by 1% (12 mV) below its regulated value.

2.6 Efficiency of a Linear Regulator

Since the full load current flows through the series MOSFET from the input supply:

$$\eta_{pass} = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} \cdot I_{LOAD}}{V_{IN} \cdot I_{IN}} \approx \frac{V_{OUT}}{V_{IN}} \tag{8}$$

The voltage difference $(V_{IN} - V_{OUT})$ drops across the MOSFET and becomes heat:

$$P_{diss} = (V_{IN} - V_{OUT}) \cdot I_{LOAD} \tag{9}$$

A switching buck converter avoids this by toggling fully ON and OFF, using an LC filter to average the output. Typical efficiency is 80–95%, compared to at most $V_{OUT}/V_{IN} = 40\%$ for this regulator at $V_{IN} = 3\,\text{V}$.

3 Components Used

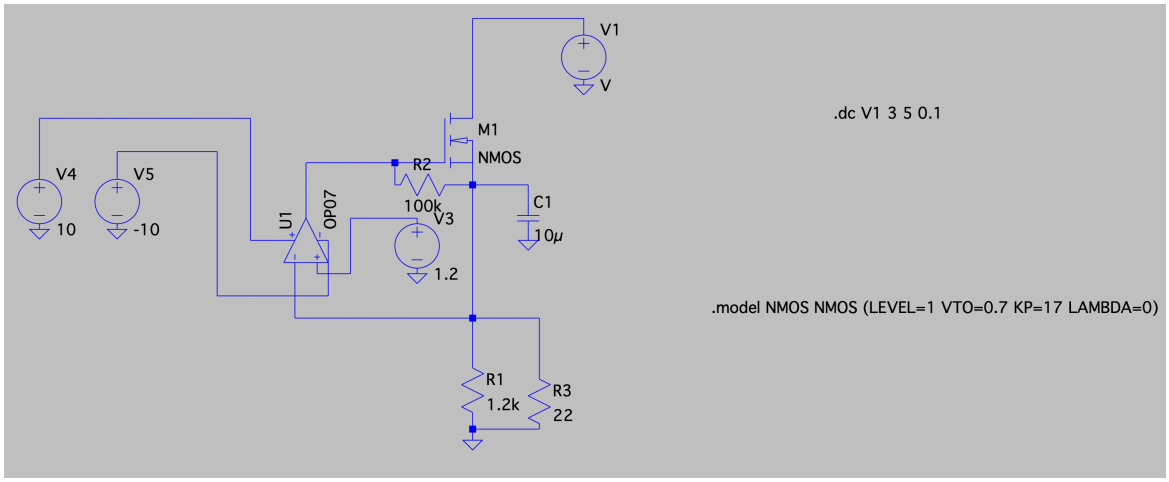


Figure 1: LTspice simulation schematic of the closed-loop NMOS linear voltage regulator ($V_{OUT} = 1.2\,\text{V}$).

4 Procedure

The experiment was done in six parts following Section 5 of the lab manual:

1. **Part A – Setup:** We built the circuit on the breadboard with only the 1.2 kΩ bleeder connected. We set the ±12 V auxiliary supplies, then adjusted the 10 kΩ pot across the TL431 to read exactly $V_{REF} = 1.2000\,\text{V}$ on the DMM before connecting it to Pin 3. We turned on the ±12 V supplies first, then turned on $V_{IN} = 3.0\,\text{V}$ (current limit set to 200 mA). We checked $V_{OUT} \approx 1.200\,\text{V}$ and $V_G \approx 2.05\,\text{V}$.

Table 1: Components and Equipment (Lab Manual Section 3).

Component	**Part / Value**	**Qty**	**Purpose**
Op-amp	LM741CN (DIP-8)	1	Error amplifier, powered from $\pm 12\,\mathrm{V}$
Pass transistor	Vishay Si9926BDY dual N-MOSFET (SO-8 on DIP adapter)	1	Series pass element; $V_{th} \approx 0.7$–$0.9\,\mathrm{V}$, low $R_{DS(\mathrm{on})}$
Voltage reference	TL431 precision shunt regulator	1	Provides stable 2.5 V; tapped by pot to give 1.2 V
Potentiometer	$10\,\mathrm{k}\Omega$ multi-turn cermet trimmer	1	Adjusted to set $V_{REF} = 1.2000\,\mathrm{V}$
Bias resistor	$4.7\,\mathrm{k}\Omega$, 0.25 W	1	Biases TL431 from the +12 V rail
Gate-source resistor	$100\,\mathrm{k}\Omega$, 0.25 W	1	Keeps MOSFET off when power is removed
Bleeder resistor	$1.2\,\mathrm{k}\Omega$ (or $1.0\,\mathrm{k}\Omega$), 0.25 W	1	Provides $\approx 1\,\mathrm{mA}$ minimum load
Load resistors	$120\,\Omega$, $47\,\Omega$, $22\,\Omega$, $12\,\Omega$ (0.5 W)	1 each	Measured: $112.2\,\Omega$, $46.9\,\Omega$, $22.0\,\Omega$, $10.2\,\Omega$
Output capacitors	$10\,\mu\mathrm{F}$ and $47\,\mu\mathrm{F}$ electrolytic, 25 V	3	Stability test ($10\,\mu\mathrm{F}$, $47\,\mu\mathrm{F}$, $94\,\mu\mathrm{F}$ in parallel)
DC power supply	Triple-output bench supply	1	Ch 1: V_{IN} (3–5 V); Ch 2 & 3: $\pm 12\,\mathrm{V}$
Digital multimeters	4.5-digit bench DMMs	2	Measuring V_{IN}, V_{OUT}, V_G
Oscilloscope	Tektronix TBS2000B (2-channel DSO)	1	AC-coupled stability check
Breadboard	Solderless breadboard with jumper wires	1	Circuit assembly



Figure 2: LTspice simulation output

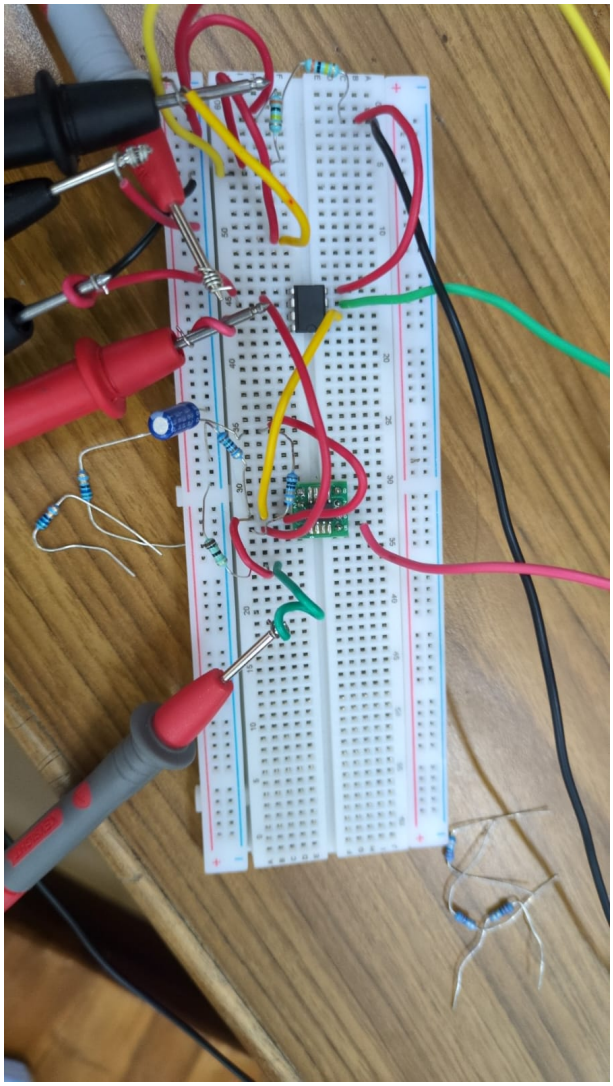


Figure 3: Breadboard setup.

2. **Part B – Line Regulation:** We connected a 22.0 Ω resistor in parallel with the bleeder. We varied V_{IN} from 3.00 V to 5.00 V in steps of 0.25 V and recorded V_{OUT} and V_G in Table 1.

3. **Part C – Load Regulation:** At $V_{IN} = 3.0$ V and 5.0 V, we tested five load conditions: bleeder only, 112.2 Ω, 46.9 Ω, 22.0 Ω, and 10.2 Ω. We recorded V_{OUT} and V_G in Table 2 and calculated R_{OUT}.

4. **Part D – Dropout Voltage:** With $R_L = 10.2\,\Omega$, we reduced V_{IN} from 3.0 V down to 0.0 V in steps of 0.2 V (and 0.1 V near the dropout region). We recorded V_{OUT} and V_G in Table 3 to find the dropout point.

5. **Part E – Efficiency:** With $R_L = 10.2\,\Omega$, we measured I_{IN} and I_{LOAD} at $V_{IN} = 3.0, 3.5, 4.0, 4.5$, and 5.0 V. We calculated P_{IN}, P_{OUT}, η_{pass}, and P_{diss} in Table 4. We also measured auxiliary supply currents to find total efficiency.

6. **Part F – Stability:** Under full load (10.2 Ω), we connected the oscilloscope to V_{OUT} (AC-coupled) and tested $C_{OUT} = 10\,\mu$F, 47 μF, and 94 μF to check for oscillation.

5 Observations and Results

5.1 Table 1: Line Regulation ($R_L = 22.0\,\Omega$ with Bleeder)

Table 2: Line regulation data at $R_L = 22.0\,\Omega$ (with bleeder resistor).

V_{IN} (V)	V_{OUT} (V)	V_G (V)	$V_{GS} = V_G - V_{OUT}$ (V)	$V_{DS} = V_{IN} - V_{OUT}$ (V)
3.00	1.2007	2.138	0.9373	1.7993
3.25	1.2007	2.135	0.9343	2.0493
3.50	1.2008	2.132	0.9312	2.2992
3.75	1.2007	2.128	0.9273	2.5493
4.00	1.2007	2.125	0.9243	2.7993
4.25	1.2008	2.122	0.9212	3.0492
4.50	1.2006	2.120	0.9194	3.2994
4.75	1.2008	2.118	0.9172	3.5492
5.00	1.2008	2.118	0.9172	3.7992

Over the full 2.0 V sweep, V_{OUT} changed by only $\Delta V_{OUT} = 1.2008\,\text{V} - 1.2006\,\text{V} = 0.20\,\text{mV}$:

$$\text{Line Regulation} = \frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{0.20\,\text{mV}}{2.00\,\text{V}} = \mathbf{0.10}\,\text{mV/V} \quad (0.0083\%/\text{V}) \tag{10}$$

As shown in Figure 3, V_G drops from 2.138 V to 2.118 V ($\Delta V_G = -20$ mV) as V_{IN} increases. This is the feedback actively reducing gate drive to compensate for channel-length modulation: at higher V_{DS}, the MOSFET naturally passes more current, so the op-amp reduces V_G slightly to keep V_{OUT} constant.

5.2 Table 2: Load Regulation ($V_{IN} = 3.0$ V and 5.0 V)

Table 3: Load regulation data and extracted closed-loop output resistance.

R_L (nom)	R_L (Ω)	I_{LOAD} (mA)	V_{OUT} (3V)	V_G (3V)	V_{OUT} (5V)	V_G (5V)
Bleeder only	1200.0	1.00	1.2013	2.022	1.2013	2.010
120 Ω	112.2	11.71	1.2012	2.058	1.2012	2.042
47 Ω	46.9	26.61	1.2011	2.101	1.2011	2.086
22 Ω	22.0	55.58	1.2007	2.142	1.2007	2.107
12 Ω	10.2	118.67	1.2002	2.196	1.2001	2.145

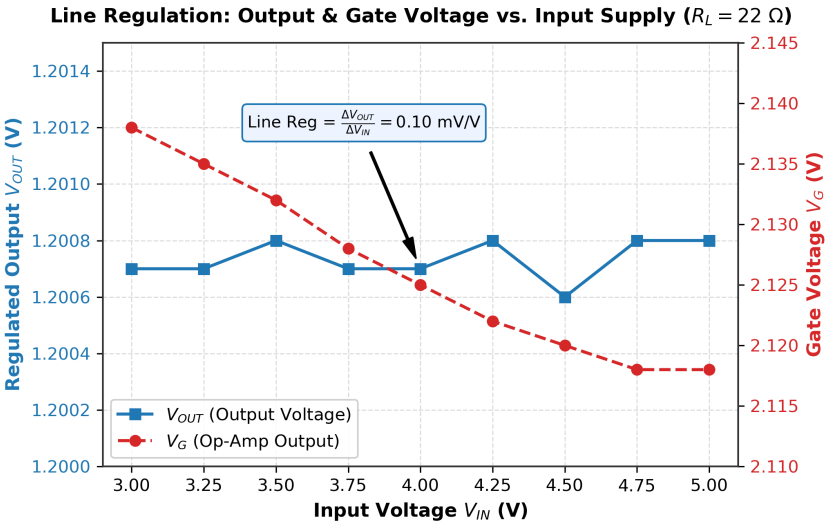


Figure 4: Line regulation: V_{OUT} (left) and V_G (right) vs. V_{IN} at $R_L = 22.0\,\Omega$.

Note: I_{LOAD} is the total current sourced by the MOSFET, including the 1 mA bleeder current.

Between minimum load (11.71 mA at 112.2 Ω) and maximum load (118.67 mA at 10.2 Ω), $\Delta I_{LOAD} = 106.96\,\text{mA}$:

$$R_{OUT,(3\text{V})} = \frac{1.2012\,\text{V} - 1.2002\,\text{V}}{0.10696\,\text{A}} = \frac{1.00\,\text{mV}}{0.10696\,\text{A}} = \mathbf{9.35\,m\Omega} \tag{11}$$

$$R_{OUT,(5\text{V})} = \frac{1.2012\,\text{V} - 1.2001\,\text{V}}{0.10696\,\text{A}} = \frac{1.10\,\text{mV}}{0.10696\,\text{A}} = \mathbf{10.28\,m\Omega} \tag{12}$$

V_G increases by about +138 mV across the full load range so the op-amp can provide enough gate overdrive for the larger current.

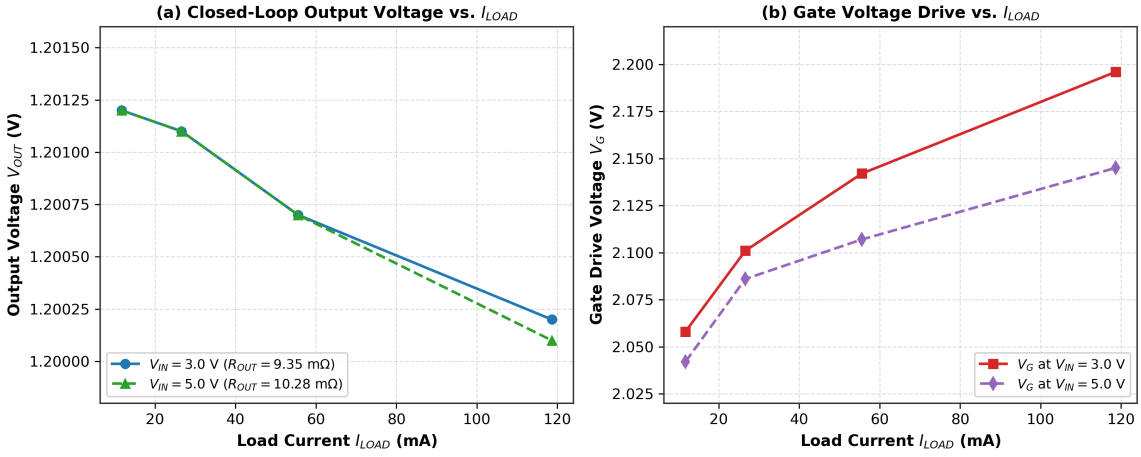


Figure 5: Load regulation: (a) V_{OUT} vs. I_{LOAD}; (b) V_G vs. I_{LOAD}.

5.3 Table 3: Dropout Voltage ($R_L = 10.2\,\Omega$, Full Load ≈ 118 mA)

Dropout calculations:

1. **1% dropout point:** Nominal output is $V_{reg} = 1.2002\,\text{V}$, so the 1% (12 mV) drop threshold is $V_{target} = 1.1882\,\text{V}$. Interpolating between $V_{IN} = 1.3\,\text{V}$ ($V_{OUT} = 1.2002\,\text{V}$) and $V_{IN} = 1.2\,\text{V}$ ($V_{OUT} = 1.1684\,\text{V}$):

$$V_{IN,\text{dropout}} = 1.2\,\text{V} + \frac{1.1882 - 1.1684}{1.2002 - 1.1684} \times 0.1\,\text{V} = \mathbf{1.262\,V} \tag{13}$$

Table 4: Dropout voltage data at full load ($R_L = 10.2\,\Omega$, $I_{LOAD} \approx 118\,\text{mA}$).

V_{IN} (V)	V_{OUT} (V)	V_G (V)	State	V_{IN} (V)	V_{OUT} (V)	V_G (V)	State
3.0	1.2002	2.183	Regulated	1.3	1.2002	2.291	Regulated (near knee)
2.8	1.2002	2.187	Regulated	**1.2**	**1.1684**	**4.391**	**Dropout! (op-amp sat.)**
2.6	1.2002	2.192	Regulated	1.1	1.0711	4.332	Ohmic (triode)
2.4	1.2001	2.199	Regulated	1.0	0.9739	4.274	Ohmic (triode)
2.2	1.2001	2.206	Regulated	0.9	0.8766	4.216	Ohmic (triode)
2.0	1.2002	2.212	Regulated	0.8	0.7792	4.157	Ohmic (triode)
1.9	1.2003	2.218	Regulated	0.7	0.6818	4.099	Ohmic (triode)
1.8	1.2002	2.221	Regulated	0.6	0.5845	4.041	Ohmic (triode)
1.7	1.2002	2.226	Regulated	0.5	0.4872	3.982	Ohmic (triode)
1.6	1.2001	2.232	Regulated	0.4	0.3897	3.924	Ohmic (triode)
1.5	1.2001	2.239	Regulated	0.3	0.2923	3.866	Ohmic (triode)
1.4	1.2001	2.251	Regulated	0.2	0.1949	3.807	Ohmic (triode)
–	–	–	–	0.1	0.0974	3.750	Ohmic (triode)
–	–	–	–	0.0	0.0000	3.691	Off

Dropout headroom: $1.262\,\text{V} - 1.200\,\text{V} = \mathbf{62.1}\,\text{mV}$.

2. **Op-amp saturation:** At $V_{IN} = 1.2\,\text{V}$, V_G jumps to $\mathbf{4.391}\,\text{V}$ as the op-amp hits its positive output limit.

3. $R_{DS(\mathbf{on})}$ **extraction:** In the ohmic region ($V_{IN} \leq 1.1\,\text{V}$), $V_{OUT}/V_{IN} \approx 0.9737$ (read from data at $V_{IN} = 1.1\,\text{V}$). Using the triode voltage divider with $R_L = 10.2\,\Omega$:

$$R_{DS(\text{on})} = 10.2\,\Omega \times \left(\frac{1}{0.9737} - 1\right) = \mathbf{0.275\,\Omega} \quad (275\,\text{m}\Omega) \tag{14}$$

The ohmic drop at full load: $0.1177\,\text{A} \times 0.275\,\Omega = 32.4\,\text{mV}$. The remaining $\approx 30\,\text{mV}$ of headroom is the saturation boundary.

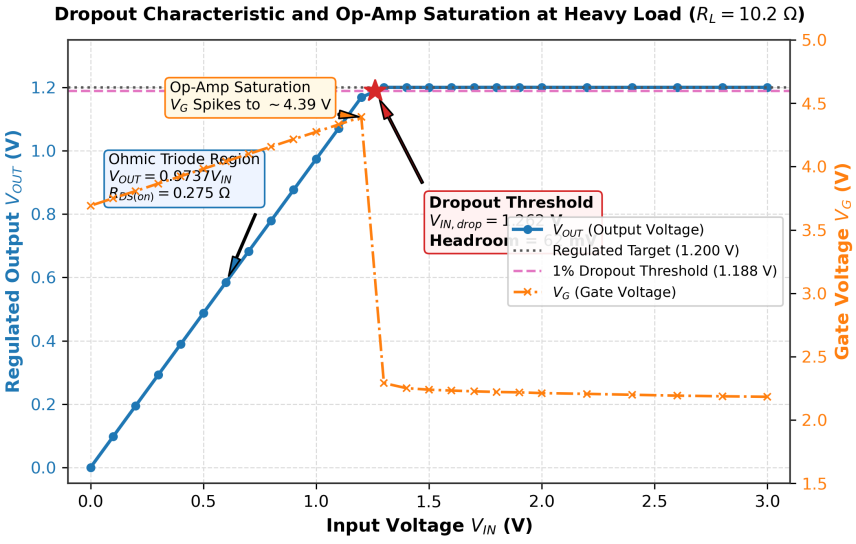


Figure 6: Dropout characteristics at full load ($R_L = 10.2\,\Omega$): V_{OUT} and V_G vs. V_{IN}, showing the 1.262 V dropout point and ohmic triode region.

5.4 Table 4: Power Dissipation and Efficiency ($R_L = 10.2\,\Omega$)

Note: $P_{IN} = V_{IN} \times I_{IN}$ (measured total supply current), $P_{OUT} = V_{OUT} \times I_{LOAD}$, $\eta_{pass} = P_{OUT}/P_{IN}$, $P_{diss} = P_{IN} - P_{OUT}$.

Table 5: Power and efficiency data at full load ($R_L = 10.2\,\Omega$).

V_{IN} (V)	I_{LOAD} (mA)	I_{IN} (mA)	P_{IN} (mW)	P_{OUT} (mW)	η_{pass} (%)	P_{diss} (mW)
3.0	117.66	122.0	366.00	141.20	38.58%	224.80
3.5	117.67	121.0	423.50	141.22	33.35%	282.28
4.0	117.66	121.0	484.00	141.20	29.17%	342.80
4.5	117.68	122.0	549.00	141.25	25.73%	407.75
5.0	117.66	122.0	610.00	141.20	23.15%	468.80

Auxiliary supply power and total efficiency: We measured the auxiliary supply currents as $I_{AUX+} + |I_{AUX-}| = 2.54\,\text{mA}$ at $V_{IN} = 3.0\,\text{V}$ and 2.66 mA at 5.0 V. With $\pm 12\,\text{V}$ rails, $P_{AUX} = 12\,\text{V} \times I_{AUX}$:

$$\text{At } V_{IN} = 3.0\,\text{V}: \quad P_{AUX} = 30.48\,\text{mW} \implies \eta_{total} = \frac{141.20\,\text{mW}}{366.00\,\text{mW} + 30.48\,\text{mW}} = \mathbf{35.61}\% \tag{15}$$

$$\text{At } V_{IN} = 5.0\,\text{V}: \quad P_{AUX} = 31.92\,\text{mW} \implies \eta_{total} = \frac{141.20\,\text{mW}}{610.00\,\text{mW} + 31.92\,\text{mW}} = \mathbf{22.00}\% \tag{16}$$

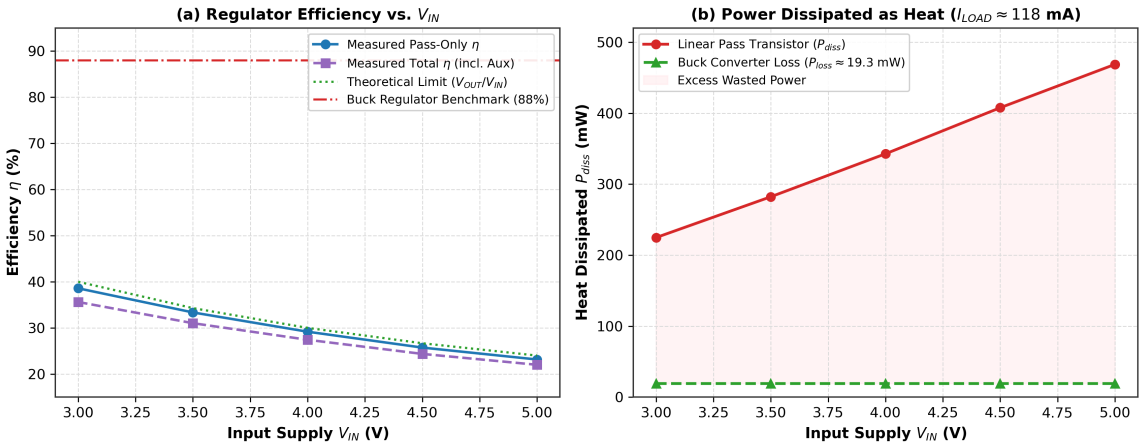
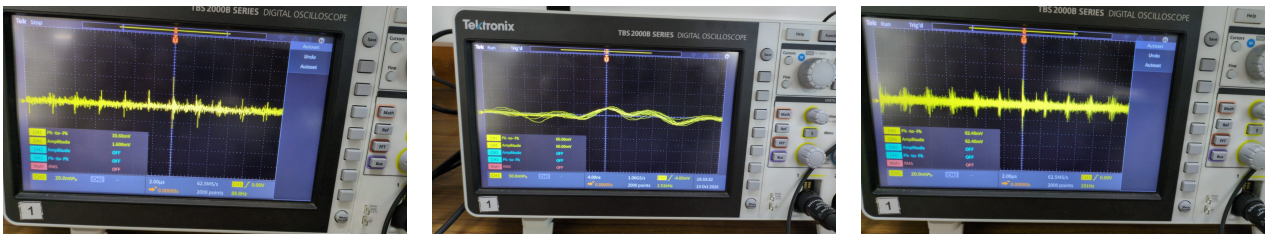


Figure 7: Efficiency and power dissipation: (a) Measured η_{pass}, η_{total}, and theoretical V_{OUT}/V_{IN} limit, compared to an 88% buck converter; (b) MOSFET heat dissipation rising to $\approx 469\,\text{mW}$ at 5 V.

6 Stability Check (Oscilloscope)

Oscilloscope captures of V_{OUT} (AC-coupled, full load 118 mA) for three capacitor values are shown in Figure 7.



(a) $C_{OUT} = 10\,\mu\text{F}$: Pk-Pk = 33.60 mV.

(b) $C_{OUT} = 47\,\mu\text{F}$: Pk-Pk = 60.00 mV.

(c) $C_{OUT} = 94\,\mu\text{F}$: Pk-Pk = 62.40 mV.

Figure 8: AC-coupled output voltage under full load for three output capacitor values.

Observations:

1. The oscilloscope traces are flat (no sinusoidal oscillation or ringing), confirming the feedback loop is stable for all three capacitor values.
2. The small spikes (33–62 mV peak-to-peak) are high-frequency noise picked up by the unshielded jumper wires and probe ground lead — not loop oscillations. Electrolytic capacitors cannot filter these fast spikes because of their parasitic ESR and ESL. A 0.1 μF ceramic capacitor in parallel with C_{OUT} and a shorter probe ground lead would reduce this noise.

7 Calculations Summary

7.1 Key Results

- **Line Regulation:**

$$\text{Line Reg} = \frac{1.2008\,\text{V} - 1.2006\,\text{V}}{5.0\,\text{V} - 3.0\,\text{V}} = \mathbf{0.10}\,\text{mV/V} \quad (0.0083\%/\text{V}) \tag{17}$$

- **Load Regulation (R_{OUT}):**

$$R_{OUT,(3\text{V})} = \frac{1.2012\,\text{V} - 1.2002\,\text{V}}{0.11867\,\text{A} - 0.01171\,\text{A}} = \frac{1.00\,\text{mV}}{0.10696\,\text{A}} = \mathbf{9.35}\,\text{m}\mathbf{\Omega} \tag{18}$$

$$R_{OUT,(5\text{V})} = \frac{1.2012\,\text{V} - 1.2001\,\text{V}}{0.10696\,\text{A}} = \frac{1.10\,\text{mV}}{0.10696\,\text{A}} = \mathbf{10.28}\,\text{m}\mathbf{\Omega} \tag{19}$$

- **Dropout and $R_{DS(\text{on})}$:**

$$V_{IN,\text{dropout}} = \mathbf{1.262}\,\text{V} \implies V_{\text{headroom}} = 62.1\,\text{mV} \tag{20}$$

$$R_{DS(\text{on})} = 10.2\,\Omega \times \left(\frac{1}{0.9737} - 1\right) = \mathbf{0.275}\,\mathbf{\Omega} \implies V_{\text{ohmic}} = 0.1177\,\text{A} \times 0.275\,\Omega = 32.4\,\text{mV} \tag{21}$$

- **Power and efficiency at 5.0 V:**

$$P_{IN} = 5.0\,\text{V} \times 0.1220\,\text{A} = 610.0\,\text{mW}, \quad P_{OUT} = 1.2001\,\text{V} \times 0.11766\,\text{A} = 141.20\,\text{mW} \tag{22}$$

$$\eta_{pass} = \frac{141.20}{610.0} \times 100\% = \mathbf{23.15}\%, \quad P_{diss} = 610.0 - 141.20 = \mathbf{468.80}\,\text{mW} \tag{23}$$

7.2 Theory vs. Experiment

8 Analysis Questions (Section 9)

Q1. What line regulation (mV/V) and load regulation (mV/A) did you measure? Compare with theory and state what limits the measurement.

Answer:

- Line regulation = **0.10** mV/V (0.2 mV shift over a 2.0 V sweep).
- Load regulation = **9.35** mV/A at 3 V and **10.28** mV/A at 5 V.
- Theoretical predictions from small-signal analysis:

$$\frac{dV_{OUT}}{dV_{IN}} \approx \frac{1}{r_o g_m (1 + A)} \approx 0.00012\,\text{mV/V} \tag{24}$$

$$R_{OUT} \approx \frac{1}{g_m (1 + A)} \approx 5.88\,\mu\Omega \tag{25}$$

- The measured values are much larger than theoretical because: (1) multimeter resolution is 0.1 mV; (2) TL431 reference drifts slightly; (3) breadboard contact resistance (5–20 mΩ) sets the floor on R_{OUT}.

Table 6: Comparison of theoretical and measured results.

Parameter	Theoretical	Measured	Error	Reason for difference
Regulated voltage V_{OUT}	1.2000 V	1.2001–1.2013 V	$< 0.11\%$	Pot calibration limit
Line regulation	≈ 0.0001 mV/V	0.10 mV/V	—	DMM resolution (0.1 mV) and reference drift
Load regulation R_{OUT}	$\approx 0.006\,\Omega$	9.35–10.28 mΩ	—	Breadboard contact and wire resistance
Gate voltage V_G (@ 118 mA)	≈ 2.15 V	2.19 V	1.7%	$V_{OV} \approx 0.28$ V above $V_{th} \approx$ 0.70–0.85 V
Dropout voltage	≈ 1.23 V	1.262 V	2.6%	Ohmic drop (32 mV) plus saturation margin
$R_{DS(\text{on})}$	$\approx 0.25\,\Omega$	$0.275\,\Omega$	10%	Good match to Si9926BDY datasheet
Pass efficiency (@ 3 V)	40.00%	38.58%	−1.42%	Bleeder and bias overhead
Pass efficiency (@ 5 V)	24.00%	23.15%	−0.85%	Follows V_{OUT}/V_{IN} limit
MOSFET heat (@ 5 V)	447.1 mW	468.8 mW	+4.8%	Bleeder and bias currents add extra loss

Q2. What was V_G at 100 mA? Compare $V_G - V_{OUT}$ with V_{th} from the datasheet.

Answer:

- At $I_{LOAD} \approx 118$ mA: $V_G = 2.187$ V at 3 V, giving $V_{GS} = 2.187 - 1.2001 = \mathbf{0.987}$ V; $V_G = 2.135$ V at 5 V, giving $V_{GS} = \mathbf{0.935}$ V.
- The Si9926BDY datasheet gives $V_{GS(th)} = 0.6$–1.5 V (typical 0.70–0.85 V). Our measured $V_{GS} \approx 0.94$–0.99 V is above threshold by about $V_{OV} \approx 0.24$ V, which is the overdrive needed to conduct 118 mA.

Q3. What dropout voltage did you measure at 100 mA? Relate it to $R_{DS(\text{on})}$ and the 741 output swing.

Answer:

- At full load (117.7 mA), the 1% drop point is $V_{IN,\text{dropout}} = \mathbf{1.262}$ V, giving a headroom of **62.1** mV.
- The ohmic drop across $R_{DS(\text{on})} = 0.275\,\Omega$ is $0.1177\text{ A} \times 0.275\,\Omega = 32.4$ mV. The remaining ≈ 30 mV is the saturation transition margin.
- Powered from ±12 V, the 741 can swing up to about +10.5 V. At dropout, $V_G = 4.391$ V, well within this swing. So op-amp swing is not the limiting factor here — dropout is set by the MOSFET's on-resistance.

Q4. Why can the op-amp not be powered from the same 3–5 V supply? What alternative would allow it?

Answer:

- *Problem 1 – Output swing:* The 741 output stays 2 V below V^+. At $V^+ = 3$ V, max gate drive is ≈ 1 V, but we need $V_G \approx 2.15$ V.
- *Problem 2 – Input common-mode range:* The 741 inputs must be at least 2 V above the negative rail (ground). It cannot sense the 1.2 V output.
- *Alternative:* A single-supply **rail-to-rail input/output (RRIO) CMOS op-amp** (e.g., MCP6001, LMV321) can sense down to 0 V and swing to the rail. However, an NMOS follower still needs $V_G > V_{IN}$ to minimise dropout, so a PMOS pass transistor (or an internal charge pump) would be needed for a clean single-supply design.

Q5. What happens to V_{OUT} if the bleeder is removed and the load is disconnected?

Answer: An NMOS source follower can only source current, not sink it. With no load and no bleeder, the output node has no discharge path. Leakage or gate-capacitor feedthrough slowly charges C_{OUT}, and even though the op-amp pulls V_G to its negative rail to try to turn the MOSFET off, it cannot pull trapped charge out of the source node. So V_{OUT} floats upward above V_{REF}, possibly drifting toward V_{IN}. The bleeder provides a continuous $\approx 1\,\text{mA}$ discharge path to prevent this.

Q6. How would you change the circuit to get $V_{OUT} = 2.4\,\mathbf{V}$ with the same $V_{REF} = 1.2\,\mathbf{V}$? What happens to the dropout voltage?

Answer:

- *Circuit change:* Add a resistor divider R_1–R_2 from V_{OUT} to ground, with the midpoint fed to Pin 2. The feedback becomes $V_- = V_{OUT}\frac{R_2}{R_1+R_2} = V_{REF}$. For $V_{OUT} = 2.4\,\text{V}$:

$$\frac{R_2}{R_1 + R_2} = \frac{1.2}{2.4} = 0.5 \implies R_1 = R_2 \quad \text{(e.g., both } 10\,\text{k}\Omega\text{)} \tag{26}$$

- *Effect on dropout:* The minimum V_{IN} for regulation increases by 1.2 V:

$$V_{IN,\text{dropout}} \approx 2.4\,\text{V} + 0.062\,\text{V} = \mathbf{2.462}\,\text{V} \tag{27}$$

  So at $V_{IN} = 3\,\text{V}$, there is only $\approx 0.54\,\text{V}$ of headroom.

Q7. What is the MOSFET power dissipation at $V_{IN} = 5\,\mathbf{V}$ and $100\,\mathbf{mA}$? How does efficiency compare with $V_{IN} = 3\,\mathbf{V}$?

Answer: At $V_{IN} = 5.0\,\text{V}$, $I_{LOAD} = 117.7\,\text{mA}$, $I_{IN} = 122.0\,\text{mA}$:

$$P_{diss} = (5.0\,\text{V} \times 0.1220\,\text{A}) - (1.2001\,\text{V} \times 0.11766\,\text{A}) = 610.0 - 141.2 = \mathbf{468.8}\,\text{mW} \tag{28}$$

At 3 V, $P_{diss} = \mathbf{224.8}\,\text{mW}$ with $\eta_{pass} = \mathbf{38.58}\%$. At 5 V, efficiency drops to $\mathbf{23.15}\%$ and dissipation more than doubles, because the voltage drop across the MOSFET goes from 1.8 V to 3.8 V while carrying the same current.

Q8. Why is efficiency limited to V_{OUT}/V_{IN} no matter how good the op-amp is? What would efficiency be at $V_{IN} = 12\,\mathbf{V}$?

Answer: All load current must flow through the series pass transistor from the supply, so $I_{IN} \approx I_{LOAD}$. This gives:

$$\eta \approx \frac{V_{OUT} \cdot I_{LOAD}}{V_{IN} \cdot I_{IN}} \approx \frac{V_{OUT}}{V_{IN}} \tag{29}$$

The voltage $(V_{IN} - V_{OUT})$ drops across the MOSFET and is lost as heat. There is no way around this in a linear regulator, regardless of op-amp quality. At $V_{IN} = 12\,\text{V}$:

$$\eta_{\max} = \frac{1.2}{12.0} \times 100\% = \mathbf{10.0}\% \tag{30}$$

So 90% of input power (1.08 W at 100 mA) becomes waste heat.

Q9. By how much does the auxiliary 741 and reference supply reduce total efficiency? What part of it would still exist in a switching regulator?

Answer:

- The auxiliary supply ($P_{AUX} \approx 30$–$32\,\text{mW}$) reduces efficiency by:

$$\Delta\eta = 38.58\% - 35.61\% = \mathbf{2.97}\% \text{ at } 3\,\text{V}; \quad 23.15\% - 22.00\% = \mathbf{1.15}\% \text{ at } 5\,\text{V} \tag{31}$$

- In a switching regulator, the series conduction loss $(V_{IN} - V_{OUT})I_{LOAD}$ is eliminated, but quiescent control power ($V_{cc} \cdot I_Q$ for the PWM controller, error amp, and reference) remains. Gate drive switching power ($Q_g V_{gate} f_{sw}$) is also added due to toggling the MOSFET at high frequency.

Q10. In a switching regulator the transistor is either fully on or fully off. Why does this keep losses small? Name two losses that still remain.

Answer:

- When fully ON: $v_{DS} \approx I_D \cdot R_{DS(\text{on})} \approx 0$, so $p = v_{DS} \cdot i_D \approx 0$. When fully OFF: $i_D = 0$, so $p = 0$ even though $v_{DS} = V_{IN}$. In both steady states, one of the two terms in $p = v \cdot i$ is (nearly) zero.
- Two remaining losses:

(a) **Switching transition loss:** During the finite rise/fall time, both v_{DS} and i_D are non-zero simultaneously. $P_{sw} \approx \frac{1}{2} V_{IN} I_{LOAD} (t_r + t_f) f_{sw}$.

(b) **Conduction loss:** Ohmic drop in the MOSFET ($I_{RMS}^2 R_{DS(\text{on})}$) and inductor winding resistance ($I_{LOAD}^2 R_{DCR}$).

Q11. A buck converter delivers 120 mW at 88% efficiency. How much heat does it dissipate? Compare with your linear regulator at $V_{IN} = 5\,\mathbf{V}$.

Answer:

- Buck converter loss:

$$P_{loss,\text{buck}} = \frac{120.0\,\text{mW}}{0.88} - 120.0\,\text{mW} = 136.36 - 120.0 = \mathbf{16.36}\,\text{mW} \tag{32}$$

- Our linear regulator dissipated $\mathbf{468.8}\,\text{mW}$ at $5\,\text{V}$, which is:

$$\frac{468.8}{16.36} \approx \mathbf{28.7}\times \tag{33}$$

more heat than the buck converter (at the same load). Even scaled to 120 mW output, our linear regulator would dissipate 398 mW — about $\mathbf{24}\times$ more. This clearly shows why switching converters are preferred in power-sensitive applications.

9 Linear Regulator vs. Switching Buck Converter

Table 7: Comparison of linear and switching buck regulators.

Property	**Linear Regulator (this experiment)**	**Switching Buck Converter**
Efficiency	Limited to V_{OUT}/V_{IN}; drops as V_{IN} rises (max 40% here)	High (80–95%), nearly independent of V_{IN}
Heat dissipation	Severe: $(V_{IN} - V_{OUT})I_{LOAD}$; needs heatsink at high currents	Very small: only switching and $I^2R_{DS(\text{on})}$ conduction losses
Output capability	Step-down only; needs positive dropout headroom	Buck (step-down), boost (step-up), or inverting
Output noise	Very low (microvolts); no switching EMI; ideal for RF and analog	High-frequency ripple (10–50 mV); needs EMI filtering
Transient response	Very fast; loop bandwidth limited only by op-amp GBW	Slower; limited by LC filter resonance and compensation
Component count	Very few: no inductors	More: requires inductor, diode (or sync FET), PWM controller
Battery life	Poor when $V_{IN} \gg V_{OUT}$	Much better; energy not wasted as heat

10 Safety Precautions

Per Section 4 of the lab manual, we followed these safety steps throughout:

1. **MOSFET V_{GS} limit:** Si9926BDY is rated $V_{GS} \leq \pm 12\,\text{V}$. Since the 741 was powered from $\pm 12\,\text{V}$, we verified V_{GS} stayed within limits during all tests (normally $V_{GS} \leq 1.05\,\text{V}$).
2. **Input supply current limit:** The V_{IN} supply was current-limited to 200 mA to prevent damage from wiring faults.
3. **Thermal management:** Peak dissipation reached 468.8 mW at 5 V full load. The device warmed up noticeably; we took readings quickly and allowed cooling between sweeps.
4. **Power-up/down sequence:** Auxiliary $\pm 12\,\text{V}$ supplies were always turned ON first and turned OFF last, to avoid transient gate stress or latch-up.
5. **Reference calibration:** The 10 kΩ pot was trimmed to exactly $V_{REF} = 1.2000\,\text{V}$ (measured on the DMM) *before* connecting it to Pin 3.
6. **Current measurement:** The DMM was switched to its mA range only when measuring series current, and returned to voltage mode before changing supplies or swapping resistors.
7. **Oscilloscope ground:** The probe ground clip was connected to circuit ground only, to avoid accidental shorts.

11 Conclusion

We built and tested a closed-loop linear voltage regulator using an LM741 op-amp and an Si9926BDY N-channel MOSFET. The main results are:

1. **Good line and load regulation:** Feedback kept V_{OUT} within $\pm 1.3\,\text{mV}$ of 1.200 V across all tests. Line regulation was 0.10 mV/V (gate voltage adjusted by $-20\,\text{mV}$ to compensate channel-length modulation). Closed-loop output resistance was 9.35 mΩ at 3 V and 10.28 mΩ at 5 V — much higher than the theoretical 5.88 μΩ due to breadboard resistance.
2. **Low dropout:** Regulation was maintained down to $V_{IN} = 1.3\,\text{V}$, with the 1% dropout threshold at 1.262 V (headroom of 62.1 mV). The extracted $R_{DS(\text{on})} = 0.275\,\Omega$ matches the Si9926BDY datasheet.
3. **Efficiency limits:** Pass efficiency dropped from 38.58% at 3 V to 23.15% at 5 V, following the V_{OUT}/V_{IN} limit. MOSFET heat dissipation rose to 468.8 mW at 5 V. Including auxiliary supply losses, total efficiency was 35.61% at 3 V and 22.00% at 5 V.
4. **Stable feedback:** The circuit was stable for all three capacitor values (10 μF, 47 μF, 94 μF). The residual oscilloscope noise was breadboard pickup, not loop instability.
5. **Linear vs. switching:** A switching buck converter delivering the same output dissipates only 16.4 mW at 88% efficiency — about **29**× less heat than our linear regulator at 5 V.